

```
In [1]: import pandas as pd  
import numpy as np
```

```
In [4]: titanic = pd.read_csv(r'C:\Users\Twinklele\OneDrive\Desktop\titanic.csv')  
titanic
```

Out[4]:

	PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare
0	1	0	3	Braund, Mr. Owen Harris	male	22.0	1	0	A/5 21171	7.25
1	2	1	1	Cumings, Mrs. John Bradley (Florence Briggs Th...	female	38.0	1	0	PC 17599	71.28
2	3	1	3	Heikkinen, Miss. Laina	female	26.0	0	0	STON/O2. 3101282	7.92
3	4	1	1	Futrelle, Mrs. Jacques Heath (Lily May Peel)	female	35.0	1	0	113803	53.10
4	5	0	3	Allen, Mr. William Henry	male	35.0	0	0	373450	8.05
...	...	...	...	...	...	...	...	...	...	...
886	887	0	2	Montvila, Rev. Juozas	male	27.0	0	0	211536	13.00
887	888	1	1	Graham, Miss. Margaret Edith	female	19.0	0	0	112053	30.00
888	889	0	3	Johnston, Miss. Catherine Helen "Carrie"	female	NaN	1	2	W./C. 6607	23.45
889	890	1	1	Behr, Mr. Karl Howell	male	26.0	0	0	111369	30.00
890	891	0	3	Dooley, Mr. Patrick	male	32.0	0	0	370376	7.73

891 rows × 12 columns



```
In [5]: titanic.tail()
```

```
Out[5]:
```

	PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare
886	887	0	2	Montvila, Rev. Juozas	male	27.0	0	0	211536	13.00
887	888	1	1	Graham, Miss. Margaret Edith	female	19.0	0	0	112053	30.00
888	889	0	3	Johnston, Miss. Catherine Helen "Carrie"	female	NaN	1	2	W./C. 6607	23.45
889	890	1	1	Behr, Mr. Karl Howell	male	26.0	0	0	111369	30.00
890	891	0	3	Dooley, Mr. Patrick	male	32.0	0	0	370376	7.75

performing data cleaning and analysis

```
In [6]: titanic.describe()
```

```
Out[6]:
```

	PassengerId	Survived	Pclass	Age	SibSp	Parch	Fare
count	891.000000	891.000000	891.000000	714.000000	891.000000	891.000000	891.000000
mean	446.000000	0.383838	2.308642	29.699118	0.523008	0.381594	32.204208
std	257.353842	0.486592	0.836071	14.526497	1.102743	0.806057	49.693429
min	1.000000	0.000000	1.000000	0.420000	0.000000	0.000000	0.000000
25%	223.500000	0.000000	2.000000	20.125000	0.000000	0.000000	7.910400
50%	446.000000	0.000000	3.000000	28.000000	0.000000	0.000000	14.454200
75%	668.500000	1.000000	3.000000	38.000000	1.000000	0.000000	31.000000
max	891.000000	1.000000	3.000000	80.000000	8.000000	6.000000	512.329200

```
In [7]: del titanic["Name"]
titanic.head()
```

Out[7]:

	PassengerId	Survived	Pclass	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked
0	1	0	3	male	22.0	1	0	A/5 21171	7.2500	NaN	
1	2	1	1	female	38.0	1	0	PC 17599	71.2833	C85	
2	3	1	3	female	26.0	0	0	STON/O2. 3101282	7.9250	NaN	
3	4	1	1	female	35.0	1	0	113803	53.1000	C123	
4	5	0	3	male	35.0	0	0	373450	8.0500	NaN	

In [8]: `del titanic["Ticket"]`
`titanic.head()`

Out[8]:

	PassengerId	Survived	Pclass	Sex	Age	SibSp	Parch	Fare	Cabin	Embarked
0	1	0	3	male	22.0	1	0	7.2500	NaN	S
1	2	1	1	female	38.0	1	0	71.2833	C85	C
2	3	1	3	female	26.0	0	0	7.9250	NaN	S
3	4	1	1	female	35.0	1	0	53.1000	C123	S
4	5	0	3	male	35.0	0	0	8.0500	NaN	S

In [9]: `del titanic["Fare"]`
`titanic.head()`

Out[9]:

	PassengerId	Survived	Pclass	Sex	Age	SibSp	Parch	Cabin	Embarked
0	1	0	3	male	22.0	1	0	NaN	S
1	2	1	1	female	38.0	1	0	C85	C
2	3	1	3	female	26.0	0	0	NaN	S
3	4	1	1	female	35.0	1	0	C123	S
4	5	0	3	male	35.0	0	0	NaN	S

In [10]: `del titanic["Cabin"]`
`titanic.head()`

```
Out[10]:
```

	PassengerId	Survived	Pclass	Sex	Age	SibSp	Parch	Embarked
0	1	0	3	male	22.0	1	0	S
1	2	1	1	female	38.0	1	0	C
2	3	1	3	female	26.0	0	0	S
3	4	1	1	female	35.0	1	0	S
4	5	0	3	male	35.0	0	0	S

```
In [11]: def getNumber(str):
          if str=="male":
              return 1
          else:
              return 2
          titanic["Gender"]=titanic["Sex"].apply(getNumber)
          titanic.head()
```

```
Out[11]:
```

	PassengerId	Survived	Pclass	Sex	Age	SibSp	Parch	Embarked	Gender
0	1	0	3	male	22.0	1	0	S	1
1	2	1	1	female	38.0	1	0	C	2
2	3	1	3	female	26.0	0	0	S	2
3	4	1	1	female	35.0	1	0	S	2
4	5	0	3	male	35.0	0	0	S	1

```
In [12]: del titanic["Sex"]
          titanic.head()
```

```
Out[12]:
```

	PassengerId	Survived	Pclass	Age	SibSp	Parch	Embarked	Gender
0	1	0	3	22.0	1	0	S	1
1	2	1	1	38.0	1	0	C	2
2	3	1	3	26.0	0	0	S	2
3	4	1	1	35.0	1	0	S	2
4	5	0	3	35.0	0	0	S	1

```
In [13]: titanic.isnull().sum()
```

```
Out[13]: PassengerId      0
Survived      0
Pclass        0
Age           177
SibSp         0
Parch         0
Embarked      2
Gender        0
dtype: int64
```

```
In [15]: means= titanic[titanic.Survived==1].Age.mean()
means
```

```
Out[15]: np.float64(28.343689655172415)
```

```
In [16]: titanic["age"]=np.where(pd.isnull(titanic.Age) & titanic["Survived"]==1 ,means, ti
titanic.head()
```

```
Out[16]:
```

	PassengerId	Survived	Pclass	Age	SibSp	Parch	Embarked	Gender	age
0	1	0	3	22.0	1	0	S	1	22.0
1	2	1	1	38.0	1	0	C	2	38.0
2	3	1	3	26.0	0	0	S	2	26.0
3	4	1	1	35.0	1	0	S	2	35.0
4	5	0	3	35.0	0	0	S	1	35.0

```
In [17]: titanic.isnull().sum()
```

```
Out[17]: PassengerId      0
Survived      0
Pclass        0
Age           177
SibSp         0
Parch         0
Embarked      2
Gender        0
age           125
dtype: int64
```

```
In [19]: means=titanic[titanic.Survived==0].Age.mean()
means
```

```
Out[19]: np.float64(30.62617924528302)
```

```
In [20]: titanic.age.fillna(means,inplace=True)
titanic.head()
```

C:\Users\Twinklele\AppData\Local\Temp\ipykernel_11604\698378622.py:1: FutureWarning: A value is trying to be set on a copy of a DataFrame or Series through chained assignment using an inplace method.
The behavior will change in pandas 3.0. This inplace method will never work because the intermediate object on which we are setting values always behaves as a copy.

For example, when doing 'df[col].method(value, inplace=True)', try using 'df.method({col: value}, inplace=True)' or df[col] = df[col].method(value) instead, to perform the operation inplace on the original object.

```
titanic.age.fillna(means,inplace=True)
```

```
Out[20]:
```

	PassengerId	Survived	Pclass	Age	SibSp	Parch	Embarked	Gender	age
0	1	0	3	22.0	1	0	S	1	22.0
1	2	1	1	38.0	1	0	C	2	38.0
2	3	1	3	26.0	0	0	S	2	26.0
3	4	1	1	35.0	1	0	S	2	35.0
4	5	0	3	35.0	0	0	S	1	35.0

```
In [21]: titanic.isnull().sum()
```

```
Out[21]: PassengerId      0
Survived      0
Pclass      0
Age      177
SibSp      0
Parch      0
Embarked      2
Gender      0
age      0
dtype: int64
```

```
In [23]: del titanic["Age"]
titanic.head()
```

```
Out[23]:
```

	PassengerId	Survived	Pclass	SibSp	Parch	Embarked	Gender	age
0	1	0	3	1	0	S	1	22.0
1	2	1	1	1	0	C	2	38.0
2	3	1	3	0	0	S	2	26.0
3	4	1	1	1	0	S	2	35.0
4	5	0	3	0	0	S	1	35.0

```
In [24]: survivedQ = titanic[titanic.Embarked == 'Q'][titanic.Survived == 1].shape[0]
survivedC = titanic[titanic.Embarked == 'C'][titanic.Survived == 1].shape[0]
survivedS = titanic[titanic.Embarked == 'S'][titanic.Survived == 1].shape[0]
```

```
print(survivedQ)
print(survivedC)
print(survivedS)
```

30
93
217

C:\Users\Twinklele\AppData\Local\Temp\ipykernel_11604\1434170922.py:1: UserWarning: Boolean Series key will be reindexed to match DataFrame index.

```
survivedQ = titanic[titanic.Embarked == 'Q'][titanic.Survived == 1].shape[0]
```

C:\Users\Twinklele\AppData\Local\Temp\ipykernel_11604\1434170922.py:2: UserWarning: Boolean Series key will be reindexed to match DataFrame index.

```
survivedC = titanic[titanic.Embarked == 'C'][titanic.Survived == 1].shape[0]
```

C:\Users\Twinklele\AppData\Local\Temp\ipykernel_11604\1434170922.py:3: UserWarning: Boolean Series key will be reindexed to match DataFrame index.

```
survivedS = titanic[titanic.Embarked == 'S'][titanic.Survived == 1].shape[0]
```

```
In [25]: survivedQ = titanic[titanic.Embarked == 'Q'][titanic.Survived == 0].shape[0]
survivedC = titanic[titanic.Embarked == 'C'][titanic.Survived == 0].shape[0]
survivedS = titanic[titanic.Embarked == 'S'][titanic.Survived == 0].shape[0]
print(survivedQ)
print(survivedC)
print(survivedS)
```

47
75
427

C:\Users\Twinklele\AppData\Local\Temp\ipykernel_11604\3240960939.py:1: UserWarning: Boolean Series key will be reindexed to match DataFrame index.

```
survivedQ = titanic[titanic.Embarked == 'Q'][titanic.Survived == 0].shape[0]
```

C:\Users\Twinklele\AppData\Local\Temp\ipykernel_11604\3240960939.py:2: UserWarning: Boolean Series key will be reindexed to match DataFrame index.

```
survivedC = titanic[titanic.Embarked == 'C'][titanic.Survived == 0].shape[0]
```

C:\Users\Twinklele\AppData\Local\Temp\ipykernel_11604\3240960939.py:3: UserWarning: Boolean Series key will be reindexed to match DataFrame index.

```
survivedS = titanic[titanic.Embarked == 'S'][titanic.Survived == 0].shape[0]
```

```
In [27]: titanic.dropna(inplace=True)
titanic.head()
```

Out[27]:

	PassengerId	Survived	Pclass	SibSp	Parch	Embarked	Gender	age
0	1	0	3	1	0	S	1	22.0
1	2	1	1	1	0	C	2	38.0
2	3	1	3	0	0	S	2	26.0
3	4	1	1	1	0	S	2	35.0
4	5	0	3	0	0	S	1	35.0

```
In [28]: titanic.isnull().sum()
```



```
Out[28]: PassengerId    0
         Survived      0
         Pclass       0
         SibSp        0
         Parch        0
         Embarked     0
         Gender       0
         age          0
         dtype: int64
```

```
In [29]: titanic.rename(columns={'age':'Age'}, inplace=True)
         titanic.head()
```

```
Out[29]:
```

	PassengerId	Survived	Pclass	SibSp	Parch	Embarked	Gender	Age
0	1	0	3	1	0	S	1	22.0
1	2	1	1	1	0	C	2	38.0
2	3	1	3	0	0	S	2	26.0
3	4	1	1	1	0	S	2	35.0
4	5	0	3	0	0	S	1	35.0

```
In [30]: titanic.rename(columns={'Gender':'Sex'}, inplace=True)
         titanic.head()
```

```
Out[30]:
```

	PassengerId	Survived	Pclass	SibSp	Parch	Embarked	Sex	Age
0	1	0	3	1	0	S	1	22.0
1	2	1	1	1	0	C	2	38.0
2	3	1	3	0	0	S	2	26.0
3	4	1	1	1	0	S	2	35.0
4	5	0	3	0	0	S	1	35.0

```
In [31]: def getEmb(str):
         if str=="S":
             return 1
         elif str=="Q":
             return 2
         else:
             return 3
         titanic["Embark"]=titanic["Embarked"].apply(getEmb)
         titanic.head()
```

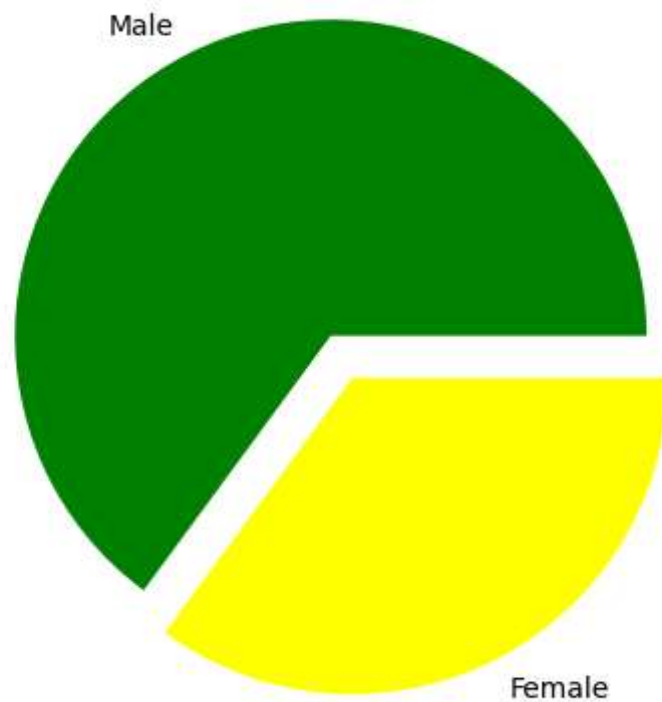
```
Out[31]:
```

	PassengerId	Survived	Pclass	SibSp	Parch	Embarked	Sex	Age	Embark
0	1	0	3	1	0	S	1	22.0	1
1	2	1	1	1	0	C	2	38.0	3
2	3	1	3	0	0	S	2	26.0	1
3	4	1	1	1	0	S	2	35.0	1
4	5	0	3	0	0	S	1	35.0	1

```
In [37]: import matplotlib.pyplot as plt
from matplotlib import style

males =(titanic['Sex']== 1).sum()
females =(titanic['Sex']== 2).sum()
print(males)
print(females)
p = [males, females]
plt.pie(p,
        labels = ['Male', 'Female'], colors = ['green', 'yellow'], explode = (0.15,
plt.axis('equal')
plt.show()
```

577
312



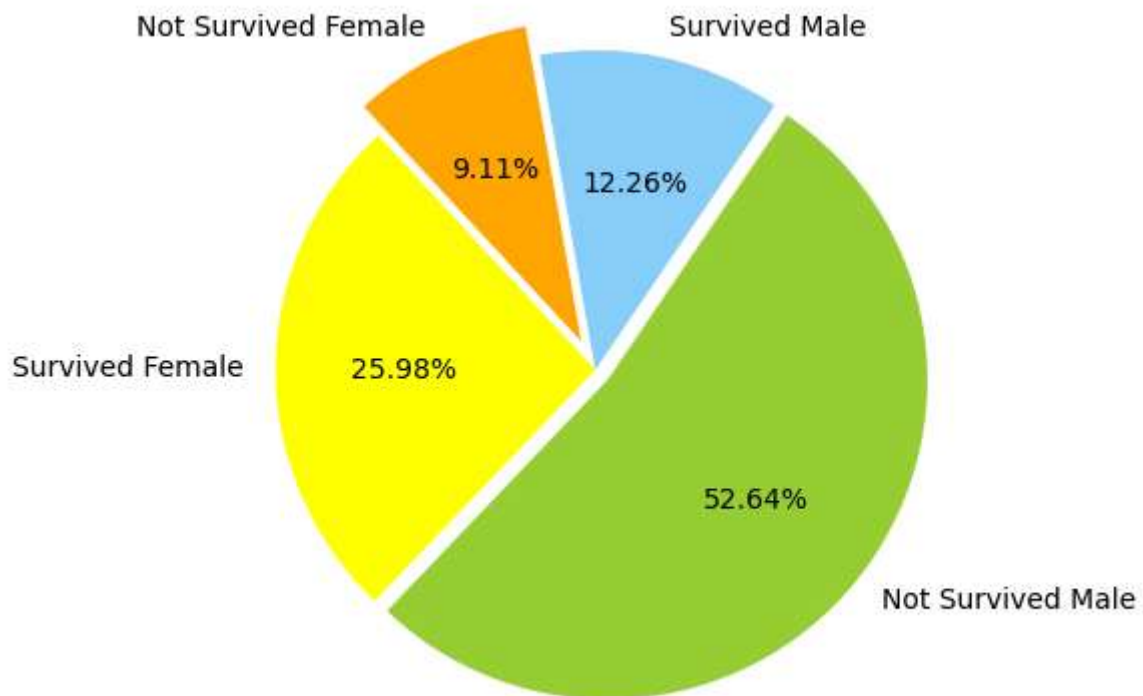
```
In [38]: MaleS=titanic[titanic.Sex==1][titanic.Survived==1].shape[0]
print(MaleS)
MaleN=titanic[titanic.Sex==1][titanic.Survived==0].shape[0]
print(MaleN)
```

```
FemaleS=titanic[titanic.Sex==2][titanic.Survived==1].shape[0]
print(FemaleS)
FemaleN=titanic[titanic.Sex==2][titanic.Survived==0].shape[0]
print(FemaleN)
```

```
109
468
231
81
```

```
C:\Users\Twinklele\AppData\Local\Temp\ipykernel_11604\1193566603.py:1: UserWarning: Boolean Series key will be reindexed to match DataFrame index.
  MaleS=titanic[titanic.Sex==1][titanic.Survived==1].shape[0]
C:\Users\Twinklele\AppData\Local\Temp\ipykernel_11604\1193566603.py:3: UserWarning: Boolean Series key will be reindexed to match DataFrame index.
  MaleN=titanic[titanic.Sex==1][titanic.Survived==0].shape[0]
C:\Users\Twinklele\AppData\Local\Temp\ipykernel_11604\1193566603.py:5: UserWarning: Boolean Series key will be reindexed to match DataFrame index.
  FemaleS=titanic[titanic.Sex==2][titanic.Survived==1].shape[0]
C:\Users\Twinklele\AppData\Local\Temp\ipykernel_11604\1193566603.py:7: UserWarning: Boolean Series key will be reindexed to match DataFrame index.
  FemaleN=titanic[titanic.Sex==2][titanic.Survived==0].shape[0]
```

```
In [40]: chart=[MaleS,MaleN,FemaleS,FemaleN]
         colors=['lightskyblue','yellowgreen','yellow','orange']
         labels=["Survived Male","Not Survived Male","Survived Female", "Not Survived Female"]
         explode=[0,0.05,0,0.1]
         plt.pie(chart,labels=labels,colors=colors,explode=explode,startangle=100,counterclockwise=True)
         plt.axis("equal")
         plt.show()
```



```
In [ ]:
```